

Engineering, Procurement and Construction Contract for the Alpilla 600 MW Combined Cycle Power Plant

Contract No. ALP-EPC-001

between

Alpilla Enerji Üretim A.Ş.

(the Owner)

and

the Consortium of

Istanbul EPC Mühendislik ve Taahhüt A.Ş. (Leader)

Imaginary Electric Company

(jointly and severally, the Contractor)

Form	Lump-sum, fixed-price, date-certain turnkey EPC contract with a consortium Contractor (joint and several liability)
Contract Price	EUR 515,000,000 (lump sum, fixed)
Date of signature	2026-10-15
Notice to Proceed	2026-11-02
Time for Completion	975 days from NTP
Governing law	Laws of the Republic of Türkiye
Status	signed

CASE STUDY. All parties, names, prices and dates are fictional and created to exercise the project environment.

Contents

- Part I - Contract Agreement
- Part II - Conditions of Contract
 - Clause 1 General Provisions
 - Clause 2 The Owner
 - Clause 3 The Engineer
 - Clause 4 The Contractor
 - Clause 5 The Consortium
 - Clause 6 Design
 - Clause 7 Staff and Labour
 - Clause 8 Plant, Materials and Workmanship
 - Clause 9 Commencement, Delays and Programme
 - Clause 10 Tests on Completion
 - Clause 11 Taking-Over
 - Clause 12 Performance Guarantees and Liquidated Damages
 - Clause 13 Defects Liability
 - Clause 14 Variations and Adjustments
 - Clause 15 Contract Price and Payment
 - Clause 16 Securities
 - Clause 17 Care of the Works and Indemnities
 - Clause 18 Insurance
 - Clause 19 Exceptional Events
 - Clause 20 Suspension and Termination
 - Clause 21 Limitation of Liability
 - Clause 22 Claims and Disputes
 - Clause 23 Miscellaneous
- Appendix A - Employer's Requirements (technical)
- Appendix B - Performance Guarantees and Liquidated Damages
- Appendix C - Schedule of Payments
- Appendix D - Scope of Works Split and Terminal Points
- Appendix E - Site and Design Data
- Appendix F - Parties and Addresses for Notices
- Appendix G - Key Dates

Part I - Contract Agreement

Article A1 Parties

This Contract is made between Alpilla Enerji Üretim A.Ş., a company incorporated in the Republic of Türkiye (the "Owner"), and the unincorporated consortium formed by Istanbul EPC Mühendislik ve Taahhüt A.Ş. (the "Leader") and Imaginary Electric Company (the "Member"), acting jointly and severally (together the "Contractor").

This is a case-study document. All parties, prices and dates are fictional.

Article A2 Recitals

- The Owner intends to develop, own and operate a natural-gas-fired combined cycle power plant of 600 MW net output on the Site on the northern Marmara coast (the "Project").
- The Owner invited tenders for the engineering, procurement and construction of the Project on a lump-sum turnkey basis and accepted the Contractor's tender as clarified during negotiations.
- The Contractor has examined the Site, the Owner's Requirements and all information made available and is satisfied as to the sufficiency of the Contract Price.

Article A3 Contract documents

The following documents form the Contract and shall be read as one. In case of conflict they take precedence in the order listed:

- this Contract Agreement;
- the Conditions of Contract (Part II);
- Appendix B - Performance Guarantees and Liquidated Damages;
- Appendix C - Schedule of Payments;
- Appendix D - Scope of Works Split and Terminal Points;
- Appendix A - Employer's Requirements (technical);
- Appendix E - Site and Design Data;
- Appendix G - Key Dates;
- the Contractor's technical proposal, to the extent it is consistent with the documents above.

Article A4 Contract Price

In consideration of the Works the Owner shall pay the Contractor the lump-sum Contract Price of EUR 515,000,000 (five hundred and fifteen million euros), comprising an offshore portion of EUR 318,000,000 and an onshore portion of EUR 197,000,000, exclusive of Value Added Tax, payable in accordance with Clause 15 and Appendix C. The Contract Price is fixed and not subject to escalation or exchange-rate adjustment.

Article A5 Time for Completion

The Contractor shall complete the Works, including passing the Tests on Completion and the Reliability Run, within 975 days from the Notice to Proceed (NTP), and shall achieve the Key Dates in Appendix G.

Article A6 Effectiveness

This Contract enters into force on signature. The Notice to Proceed shall be issued within 21 days after the last of the following conditions precedent has been fulfilled: (a) delivery of the Performance Security and the Advance Payment Guarantee; (b) delivery of the parent company guarantees of both consortium members; (c) financial close of the Owner's project financing; and (d) the positive EIA decision remaining valid and unchallenged.

Article A7 Signatures

In witness whereof the Parties have caused this Contract to be executed by their duly authorised representatives on the date stated on the cover page.

For the Owner Alpilla Enerji Üretim A.Ş.	For the Contractor (Leader) Istanbul EPC Mühendislik ve Taahhüt A.Ş.	For the Contractor (Member) Imaginary Electric Company
Name:	Name:	Name:
Title:	Title:	Title:
Signature:	Signature:	Signature:
Date:	Date:	Date:

Part II - Conditions of Contract

1 General Provisions

1.1 Definitions

In the Contract the following words have the meanings stated, unless the context requires otherwise:

- "Works" means the permanent works and temporary works required to deliver the complete power plant described in the Employer's Requirements.
- "Site" means the land made available by the Owner as described in ER-02.01, including the marine areas for intake and outfall.
- "Engineer" means the person appointed by the Owner to act as the Owner's engineer and notified to the Contractor.
- "Taking-Over" means the stage at which the Owner takes over the Works under Clause 11.
- "Defects Notification Period" means the period stated in Sub-Clause 13.1.
- "Site Reference Conditions" means the conditions stated in ER-02.03.
- "Good Industry Practice" means the skill, diligence and foresight reasonably expected of an experienced international contractor executing works of similar size and complexity.
- "Owner's Database" means the engineering database designated by the Owner as the single source of project data.

1.2 Interpretation

Words in the singular include the plural and vice versa. Headings do not affect interpretation. "Including" means including without limitation. A reference to a law, code or standard is to the version in force at the Base Date (thirty days before the tender submission date), unless stated otherwise.

1.3 Priority of documents

The documents forming the Contract are mutually explanatory. Ambiguities or discrepancies shall be resolved by the order of precedence in Article A3. Where a discrepancy exists within the Employer's Requirements, the more stringent requirement applies. The Contractor shall notify any discrepancy it discovers without delay.

1.4 Law and language

The Contract is governed by the laws of the Republic of Türkiye. The ruling language is English. Communications with Turkish authorities shall be in Turkish, prepared by the Contractor where the Contractor is responsible for the underlying permit or approval.

1.5 Communications

Notices, consents, approvals and instructions shall be in writing and delivered through the project document management system or by hand to the addresses in Appendix F. A notice is effective when received. Informal communications, including meeting minutes, do not constitute a notice unless expressly stated.

1.6 Confidentiality and data

Each Party shall keep the other Party's confidential information confidential and use it only for the Project. The Owner may use all design data, documents and database content delivered by the Contractor for the operation, maintenance, repair, extension and refinancing of the Project.

2 The Owner

2.1 Access to the Site

The Owner shall give the Contractor access to and possession of the Site, levelled to +15.0 m, within 14 days after NTP, and access to the marine areas once the relevant permits are obtained under Sub-Clause 2.2.

2.2 Permits

The Owner shall obtain and maintain the EIA decision, the generation licence, the grid connection agreement and the land-use permits. The Contractor shall obtain all other permits for the design, construction and testing of the Works and shall provide all technical inputs the Owner needs for the Owner's permits.

2.3 Owner's financial arrangements

The Owner shall, within 28 days after receiving a reasonable request, provide reasonable evidence that financial arrangements are in place to pay the Contract Price.

2.4 Owner-furnished items and utilities

The Owner shall provide the items allocated to it in Appendix D, including natural gas for commissioning and tests, grid access for back-feed and test energy, and utilities to the Site boundary, at the times required by the programme accepted under Sub-Clause 9.3.

3 The Engineer

3.1 Role of the Engineer

The Engineer acts for the Owner in reviewing documents, inspecting and witnessing tests, certifying payments and determining claims. The Engineer has no authority to amend the Contract. Any approval, review or inspection by the Engineer does not relieve the Contractor of any responsibility.

3.2 Instructions

The Contractor shall comply with instructions of the Engineer given in accordance with the Contract. If the Contractor considers that an instruction constitutes a Variation, it shall notify the Engineer within 14 days before commencing the work, unless the instruction concerns safety.

4 The Contractor

4.1 General obligations

The Contractor shall execute the Works in accordance with the Contract, Good Industry Practice and applicable laws, so that on completion the Works are fit for the purposes defined in the Employer's Requirements. The Contractor shall provide all design, plant, materials, labour, equipment and services required, whether temporary or permanent.

4.2 Performance Security

The Contractor shall deliver an unconditional, irrevocable on-demand bank guarantee from a bank acceptable to the Owner for 10 % of the Contract Price, valid until the issue of the Performance Certificate, reducing to 5 % at Taking-Over.

4.3 Contractor's representative

The Contractor shall appoint a Project Director resident on Site with full authority to act for the Contractor. Replacement requires the Owner's prior consent, not to be unreasonably withheld.

4.4 Subcontractors and vendors

The Contractor shall not subcontract the whole of the Works. Subcontracting of the items listed as major equipment in ER-01.08 and of civil works above EUR 5,000,000 requires the Owner's prior consent. The Contractor remains responsible for all subcontractors.

4.5 Site data

The Owner has made available the Site data listed in Appendix E. The Contractor is responsible for interpreting that data and for any additional investigation. The Contractor is entitled to relief only for physical

conditions that an experienced contractor could not have foreseen at the Base Date after reasonable investigation.

4.6 Health, safety and environment

The Contractor shall comply with all HSE laws, the EIA conditions and the Owner's HSE requirements, and shall stop any unsafe work. Serious incidents shall be reported to the Owner within 24 hours.

4.7 Quality management

The Contractor shall implement the quality system required by ER-15 and give the Engineer access to all places of manufacture. Compliance with the quality system does not relieve the Contractor of any obligation.

4.8 Local content

The Contractor shall use reasonable endeavours to procure goods and services in Türkiye where competitive in quality, price and delivery, and shall report the local content achieved each quarter.

4.9 Engineering data

The Contractor shall deliver all engineering data into the Owner's Database in accordance with ER-01.06, ER-01.10 and ER-18.04. Data in the Owner's Database prevails over any conflicting copy.

5 The Consortium

5.1 Joint and several liability

The Leader and the Member are jointly and severally liable to the Owner for the performance of all obligations of the Contractor. The Owner may claim against either or both of them.

5.2 Authority of the Leader

The Leader is authorised to bind the Contractor, to receive notices and instructions and to receive payments for the Contractor. Notices given to the Leader are deemed given to both members.

5.3 Composition

The composition of the consortium and the allocation of work set out in Appendix D shall not be changed without the Owner's prior consent.

5.4 Internal arrangements

The consortium agreement between the Leader and the Member governs only their internal relationship. No provision of it limits their liability towards the Owner.

5.5 Insolvency of a member

If a member becomes insolvent or ceases to perform, the other member shall complete the Works without additional cost to the Owner, with a replacement acceptable to the Owner where required for the power island.

6 Design

6.1 General design obligations

The Contractor shall carry out and be responsible for the design of the Works, performed by qualified engineers. The design shall comply with the Employer's Requirements and the laws and codes stated therein.

6.2 Contractor's documents and review

The Contractor shall submit the documents listed in the master document register for review. The Engineer shall respond within the review period in ER-17.01 with one of: code 1 (no comments), code 2 (proceed

subject to comments), code 3 (revise and resubmit) or code 4 (for information). Work shall not start on items subject to review until code 1 or 2 has been obtained.

6.3 Errors in the Contractor's documents

Errors, omissions or ambiguities in the Contractor's documents shall be corrected at the Contractor's cost, notwithstanding any review.

6.4 As-built documents and database

The Contractor shall deliver as-built documents, the 3D model and the final engineering database in accordance with ER-17.03 as a condition of the Performance Certificate.

6.5 Intellectual property

Copyright and other intellectual property in the Contractor's documents remain with the Contractor or its suppliers. The Owner receives a non-exclusive, perpetual, royalty-free licence to use them for the Project. Proprietary source code of turbine control systems remains with the supplier and is placed in escrow.

7 Staff and Labour

7.1 Engagement of staff

The Contractor shall engage staff and labour in compliance with Turkish labour law, social security law and work permit requirements, and shall pay wages and benefits not lower than those established for the trade in the region.

7.2 Key personnel

The Contractor shall provide the key personnel named in its tender (project director, engineering manager, construction manager, commissioning manager, HSE manager, quality manager) and shall not replace them without the Owner's consent.

7.3 Accommodation and welfare

The Contractor shall provide accommodation, transport and welfare facilities for its personnel in accordance with applicable standards and the IFC/EBRD worker accommodation guidance.

8 Plant, Materials and Workmanship

8.1 Manner of execution

Plant and materials shall be new, of the quality specified and suitable for the intended purpose. Workmanship shall be carried out in a proper and workmanlike manner.

8.2 Inspection

The Engineer may inspect, examine and test materials and workmanship during manufacture and construction at any reasonable time. The Contractor shall give notice when work is ready and before it is covered up.

8.3 Testing

The Contractor shall perform the tests specified in the Contract, including FATs listed in ER-15.03. The Engineer shall be given at least 15 days' notice of witness points.

8.4 Rejection and remedial work

If plant, materials or workmanship are found defective, the Engineer may reject them and the Contractor shall promptly remedy the defect and ensure that the rejected item complies with the Contract.

8.5 Ownership of plant and materials

Title in plant and materials passes to the Owner on the earlier of their delivery to Site or payment for them, free of liens and encumbrances. Risk remains with the Contractor under Clause 17 until Taking-Over.

9 Commencement, Delays and Programme

9.1 Commencement

The Contractor shall commence the Works on the NTP date and proceed with due expedition and without delay.

9.2 Time for Completion and Key Dates

The Contractor shall complete the Works within the Time for Completion (975 days from NTP) and achieve the Key Dates in Appendix G, in particular first fire, first synchronisation in combined cycle and Taking-Over.

9.3 Programme

The Contractor shall submit a Level 3 CPM programme within 28 days after NTP in accordance with ER-17.02 and update it monthly. The accepted programme does not relieve the Contractor of any obligation.

9.4 Extension of Time

The Contractor is entitled to an extension of the Time for Completion if and to the extent that completion is or will be delayed by: a Variation; an Exceptional Event under Clause 19; unforeseeable physical conditions under Sub-Clause 4.5; a delay caused by the Owner or its other contractors; or a change in law under Sub-Clause 14.4; in each case subject to Clause 22. Concurrent delays caused by the Contractor are taken into account.

9.5 Delay damages

If the Contractor fails to achieve Taking-Over within the Time for Completion, it shall pay delay damages of 0.1 % of the Contract Price per day of delay. Delay damages are capped at 10 % of the Contract Price. If first synchronisation in combined cycle is delayed by more than 60 days beyond its Key Date, interim delay damages of 0.05 % per day apply, to be credited against final delay damages.

9.6 Rate of progress

If progress is too slow to complete within the Time for Completion, the Engineer may instruct the Contractor to submit a revised programme and to accelerate at the Contractor's cost.

10 Tests on Completion

10.1 Contractor's obligations

The Contractor shall carry out the Tests on Completion described in ER-16 after providing the as-built documents and O&M manuals required for testing. Test procedures shall be submitted 90 days before the tests.

10.2 Delayed tests

If the Owner unduly delays the Tests on Completion for more than 30 days, the Contractor may give notice and the Works are deemed to be taken over, subject to the tests being carried out when possible.

10.3 Retesting

If the Works fail any Test on Completion, the Contractor shall remedy the failure and repeat the test at its cost, including the cost of fuel above the quantities allowed for the first test.

10.4 Failure to pass Tests on Completion

If the Works fail to pass the performance tests after the rectification period in Sub-Clause 12.5, the consequences in Clause 12 apply.

11 Taking-Over

11.1 Conditions for Taking-Over

The Engineer shall issue the Taking-Over Certificate when all of the following are fulfilled: the Works are complete except for minor punch items not affecting safe operation; all Tests on Completion have been passed or performance liquidated damages have been paid; the Reliability Run has been passed; emission and environmental guarantees are met; the required permits for operation are available; and training has been completed.

11.2 No partial taking-over

The Works shall be taken over as a whole. Operation of the gas turbine in open cycle during commissioning does not constitute taking-over, and the energy produced belongs to the Owner.

11.3 Punch list

Punch items listed at Taking-Over shall be completed within 90 days. The Owner may withhold twice the estimated value of open punch items from the Taking-Over milestone payment.

12 Performance Guarantees and Liquidated Damages

12.1 Guarantees

The Contractor guarantees that the Works achieve the values in Appendix B when tested in accordance with the Employer's Requirements.

12.2 Performance liquidated damages

If the Works fail to achieve a guaranteed value for which a liquidated damages rate is stated in Appendix B, the Contractor shall pay liquidated damages at that rate, pro rata for fractions.

12.3 Minimum acceptance levels

If the net output is below 97 % of the guaranteed value or the net heat rate is above 103 % of the guaranteed value after the rectification period, the Owner may reject the Works, in which case Sub-Clause 20.2 applies.

12.4 Must-meet guarantees

The emission, noise near-field and environmental guarantees marked "must rectify" in Appendix B are absolute. The Contractor shall rectify the Works until they are met; Taking-Over shall not be issued before compliance.

12.5 Rectification period

The Contractor may, within 180 days after the first performance test, carry out modifications and repeat the test at its cost. The Owner may operate the plant commercially during this period.

12.6 Caps

Performance liquidated damages are capped at 15 % of the Contract Price. The aggregate of delay and performance liquidated damages is capped at 20 % of the Contract Price.

12.7 Nature of liquidated damages

The Parties agree that the liquidated damages are a genuine pre-estimate of the Owner's loss and are the Owner's sole financial remedy for the failure concerned, without prejudice to the Owner's rights under Sub-Clauses 12.3, 12.4 and 20.2.

13 Defects Liability

13.1 Defects Notification Period

The Defects Notification Period is 24 months from Taking-Over. For marine works, civil structures and buildings it is 36 months.

13.2 Remedying defects

The Contractor shall remedy any defect notified during the Defects Notification Period at its cost within the time reasonably stated by the Owner, including dismantling, transport and re-installation. If the Contractor fails to do so, the Owner may carry out the work at the Contractor's cost.

13.3 Extension of the Defects Notification Period

The Defects Notification Period of a part of the Works is extended by the period during which that part cannot be used because of a defect, and restarts for 12 months for replaced parts, up to a maximum of 48 months from Taking-Over.

13.4 Serial defects

If the same defect occurs in three or more identical components, it is deemed a serial defect and the Contractor shall replace or modify all identical components, including those not yet failed.

13.5 Performance Certificate

The Engineer shall issue the Performance Certificate within 28 days after the latest expiry of the Defects Notification Periods, when all defects have been remedied and the as-built documentation has been delivered.

14 Variations and Adjustments

14.1 Right to vary

The Owner may instruct Variations at any time before Taking-Over. The Contractor shall execute them unless it promptly demonstrates that it cannot readily obtain the required goods or that the Variation would affect safety or the guarantees.

14.2 Value engineering

The Contractor may propose changes that reduce cost, accelerate completion or improve efficiency. If accepted, the net saving is shared equally between the Parties.

14.3 Variation procedure

Before a Variation is instructed, the Engineer may request a proposal. The Contractor shall respond within 28 days with a description, programme impact and price based on the rates in the Contract or, where none apply, on reasonable cost plus 10 % overhead and profit.

14.4 Change in law

The Contract Price and Time for Completion shall be adjusted for changes in Turkish law after the Base Date that affect the Contractor's cost or time, excluding changes to corporate income tax of the Contractor.

15 Contract Price and Payment

15.1 Contract Price

The Contract Price is the lump sum stated in Article A4. It includes all duties, taxes and fees payable by the Contractor except Value Added Tax, which the Owner pays in addition where applicable, and customs duties on imported permanent plant, which are exempted under the Owner's investment incentive certificate.

15.2 Advance payment

The Owner shall pay an advance of 10 % of the Contract Price within 30 days after receiving the Advance Payment Guarantee and an invoice. The advance is recovered pro rata from milestone payments MS-02 to MS-10.

15.3 Milestone payments

The Contractor may invoice each milestone in Appendix C when it has been achieved and certified by the Engineer. No payment is due for partially achieved milestones.

15.4 Retention

The Owner shall retain 5 % of each milestone payment. Half of the retention is released at Taking-Over and half at the Performance Certificate; alternatively the retention may be released against a Retention Guarantee of equal value.

15.5 Time for payment

The Owner shall pay each certified amount within 45 days after receiving the Contractor's invoice with supporting documents.

15.6 Delayed payment

Late payments bear interest at EURIBOR (3 months) plus 3 % per annum, calculated daily.

15.7 Currency

Payments are made in euros to the accounts designated by the Leader for the offshore and onshore portions respectively.

16 Securities

16.1 Securities to be provided

The Contractor shall provide: the Performance Security (Sub-Clause 4.2); an Advance Payment Guarantee equal to the advance, reducing as it is recovered; a Retention Guarantee if retention is released early; and a parent company guarantee from the ultimate parent of each consortium member covering its obligations under the Contract.

16.2 Claims under securities

The Owner may claim under a security only for amounts to which it is entitled under the Contract and shall give the Contractor 7 days' notice before a claim, except where the security is about to expire.

17 Care of the Works and Indemnities

17.1 Care of the Works

The Contractor bears full responsibility for the care of the Works from NTP until Taking-Over, and thereafter for any outstanding work.

17.2 Owner's risks

The Owner bears the risk of loss or damage caused by: war, hostilities, rebellion or civil war; ionising radiation not caused by the Contractor; pressure waves from aircraft; use or occupation of part of the Works by the Owner; and design provided by the Owner.

17.3 Indemnities

The Contractor shall indemnify the Owner against claims for death, injury, loss or damage to property arising from the execution of the Works, except to the extent caused by the negligence of the Owner. The Owner shall indemnify the Contractor against claims arising from the Owner's risks.

18 Insurance

18.1 Construction and erection all risks

The Contractor shall procure a construction and erection all risks (CEAR) policy covering the Works for their full replacement value, including testing and commissioning, with the Owner and its lenders as co-insured and the lenders as loss payee.

18.2 Third party liability

The Contractor shall maintain third party liability insurance of not less than EUR 50,000,000 per occurrence.

18.3 Marine cargo

The Contractor shall insure all plant shipped to Site under a marine cargo policy on a warehouse-to-site basis.

18.4 Delay in start-up

The Owner may procure delay in start-up (DSU) insurance. The Contractor shall cooperate with the insurers and comply with their reasonable risk-management recommendations.

19 Exceptional Events

19.1 Definition

An Exceptional Event is an event beyond a Party's control which it could not reasonably have provided against, could not reasonably have avoided or overcome, and which is not substantially attributable to the other Party, including war, terrorism, epidemics declared by public authorities, natural catastrophes such as earthquakes of intensity exceeding the design basis, and nationwide strikes. Earthquakes within the design basis of the Works are not Exceptional Events.

19.2 Notice

The affected Party shall give notice within 14 days after becoming aware of the Exceptional Event and shall use all reasonable endeavours to minimise the delay.

19.3 Consequences

The Contractor is entitled to an extension of time for delays caused by an Exceptional Event and, for war, terrorism and riots in Türkiye, to its reasonable costs.

19.4 Prolonged Exceptional Event

If the execution of substantially all the Works is prevented for a continuous period of 180 days, either Party may terminate the Contract by 30 days' notice. The Owner shall then pay for work done and the cost of demobilisation.

20 Suspension and Termination

20.1 Suspension by the Owner

The Owner may suspend the Works at any time. If the suspension exceeds 84 days not caused by the Contractor, the Contractor is entitled to an extension of time and costs, and may treat the suspension as an omission of the affected part after a further 28 days' notice.

20.2 Termination for Contractor's default

The Owner may terminate the Contract by 14 days' notice if the Contractor: fails to provide the Performance Security; abandons the Works; fails without reasonable excuse to proceed; reaches the cap of delay damages; fails the minimum acceptance levels in Sub-Clause 12.3; becomes insolvent (both members); or commits bribery. The Owner may then complete the Works through others and recover the additional cost, and may reject the Works and recover all amounts paid where the plant is incapable of commercial use.

20.3 Termination for convenience

The Owner may terminate the Contract for convenience by 56 days' notice. The Owner shall then pay for work done, goods ordered, demobilisation costs and 5 % of the value of the work not executed as compensation for loss of profit.

20.4 Termination by the Contractor

The Contractor may terminate the Contract if the Owner fails to pay a certified amount within 90 days after its due date, or suspends the Works for more than 180 days without cause, after giving 14 days' notice to remedy.

21 Limitation of Liability

21.1 Total liability

The total liability of the Contractor under or in connection with the Contract shall not exceed 100 % of the Contract Price.

21.2 Indirect loss

Neither Party is liable to the other for loss of use, loss of profit, loss of contract or any indirect or consequential loss, except as expressly provided by the liquidated damages in Clauses 9 and 12.

21.3 Exceptions

The limitations in this Clause do not apply in case of fraud, gross negligence, wilful misconduct, abandonment of the Works, or to the indemnities for death and personal injury and third-party property damage.

22 Claims and Disputes

22.1 Claims

A Party that considers itself entitled to additional time or payment shall give notice within 28 days after becoming aware of the event, and a fully detailed claim within 84 days. Failure to give timely notice results in the loss of the entitlement, except where the other Party was not prejudiced.

22.2 Dispute Avoidance/Adjudication Board

A standing three-member Dispute Avoidance/Adjudication Board shall be appointed within 42 days after NTP. It visits the Site every four months, assists in avoiding disputes and gives decisions on referred disputes within 84 days. Decisions are binding unless and until revised by arbitration.

22.3 Amicable settlement

Before arbitration the Parties shall attempt to settle the dispute through meetings of senior executives within 28 days after a notice of dissatisfaction with a Board decision.

22.4 Arbitration

Disputes not settled shall be finally resolved under the Rules of Arbitration of the International Chamber of Commerce by three arbitrators, seated in Istanbul, in English. The Works shall continue during any dispute.

23 Miscellaneous

23.1 Anti-bribery and compliance

Each Party shall comply with applicable anti-bribery laws, export control and sanctions regulations. Breach by the Contractor entitles the Owner to terminate under Sub-Clause 20.2.

23.2 Assignment

Neither Party may assign the Contract without the other's consent, except that the Owner may assign its rights by way of security to its lenders. The Contractor shall sign a direct agreement with the lenders in customary form.

23.3 Entire agreement and amendments

The Contract is the entire agreement between the Parties about its subject matter. Amendments are valid only if in writing and signed by both Parties; each amendment shall be recorded in the Owner's Database.

23.4 Severability

If a provision is held invalid, the remaining provisions remain in force and the Parties shall replace the invalid provision with a valid one of the closest economic effect.

Appendix A - Employer's Requirements (technical)

The requirements below are mandatory ("shall"). Each states its verification method and, where applicable, the acceptance criterion and the related guarantee in Appendix B.

ER-01 General requirements and plant description

ER-01.01 Purpose of the Works

The Contractor shall design, engineer, procure, manufacture, deliver, construct, erect, commission, test and hand over a complete, safe and fully operational natural-gas-fired combined cycle power plant of 600 MW net output at Site Reference Conditions (the Works), ready for commercial operation in the Turkish electricity market.

Discipline: general | Verification: review | Acceptance: Taking-Over Certificate issued

ER-01.02 Plant configuration

The plant shall be a 1x1 multi-shaft combined cycle block comprising one heavy-duty H-class (or equivalent advanced class) gas turbine with its own generator, one triple-pressure reheat heat recovery steam generator (HRSG) with selective catalytic reduction (SCR) and CO catalyst, and one reheat condensing steam turbine with its own generator, cooled by once-through seawater.

Discipline: process | Verification: review | Acceptance: Configuration as described in the approved Basic Design Package

ER-01.03 Design life and operating regime

All equipment and structures shall be designed for an economic life of 30 years, 8,000 equivalent operating hours per year and up to 250 starts per year (of which up to 50 cold, 100 warm and 100 hot starts), including daily cycling and load following.

Discipline: general | Verification: analysis | Acceptance: Design life documented in design reports and OEM statements

ER-01.04 Turnkey completeness

The Works shall include everything necessary for safe, reliable and efficient operation whether or not expressly mentioned in the Contract (fit-for-purpose obligation), except the items expressly allocated to the Owner in Appendix D (scope split).

Discipline: general | Verification: review

ER-01.05 Units and language

SI units shall be used throughout (pressure in bar(g)/bar(a), temperature in degC). All documents, drawings and labels shall be in English; operator-facing labels and safety signs shall be bilingual English/Turkish.

Discipline: general | Verification: review

ER-01.06 Plant identification system

All systems, equipment, instruments, cables and documents shall be identified according to KKS (VGB-B 105/106). The tag data shall be classified according to the AVEVA class library adopted by the Owner and delivered through the Owner's engineering database.

Discipline: general | Verification: review | Acceptance: 100 % of delivered tags classified and validated in the Owner's database

ER-01.07 Codes and standards hierarchy

The Works shall comply with (in order of precedence): Turkish laws and regulations; the Grid Code of the TSO; the Contract; the codes and standards listed in Appendix A.19; and good industry practice. Where codes conflict the more stringent requirement applies.

Discipline: general | Verification: review

ER-01.08 Proven technology

Major equipment (GT, ST, generators, HRSG, GSU transformers) shall be of a design with at least two units of the same frame in commercial operation for not less than 8,000 fired hours each at the date of the Contract, or supported by a validation programme accepted by the Owner.

Discipline: general | Verification: review | Acceptance: Reference list accepted by the Owner

ER-01.09 Design responsibility

The Contractor is fully responsible for the design, including the verification of all Owner-provided data. Review or approval of documents by the Owner or the Engineer does not relieve the Contractor of any obligation.

Discipline: general | Verification: review

ER-01.10 Engineering database and data handover

The Contractor shall deliver all engineering data (equipment, lines, instruments, cables, documents, design parameters) in the Owner's engineering database format, kept current at each document issue, and hand over native CAD/3D models (Plant 3D or E3D) and IFC4 exports.

Discipline: general | Verification: review | Acceptance: Data handover accepted at Taking-Over

ER-02 Site conditions and design data

ER-02.01 Site location

The Site is located on the northern Marmara coast of Türkiye (coastal industrial zone), approximately 12 hectares, levelled to +15.0 m above mean sea level, with direct access to the sea for cooling water intake and outfall.

Discipline: civil | Verification: inspection

ER-02.02 Ambient design range

Equipment shall operate at full capability between ambient dry-bulb temperatures of -8 degC and +40 degC, relative humidity 20 % to 100 %, and shall be designed for survival (not operation) between -15 degC and +45 degC.

Discipline: process | Verification: analysis

ER-02.03 Site Reference Conditions (SRC)

Guarantees refer to: ambient dry-bulb temperature 15.0 degC; relative humidity 70 %; barometric pressure 1011.5 mbar; seawater temperature at intake 16.0 degC; natural gas per ER-03.02 reference composition; grid frequency 50.0 Hz; power factor 0.85 lagging.

Discipline: process | Verification: review

ER-02.04 Seawater conditions

Seawater temperature range 6 degC to 27 degC at the intake; salinity 22 g/l to 26 g/l; suspended solids up to 50 mg/l during storms; presence of jellyfish and mussel larvae shall be considered in screen and biofouling design.

Discipline: process | Verification: analysis

ER-02.05 Wind, snow and rain

Basic wind speed 30 m/s (10 min mean, 10 m height, 50-year return); ground snow load 0.75 kN/m²; design rainfall intensity 110 mm/h (10 min, 25-year return) for drainage design.

Discipline: civil | Verification: analysis

ER-02.06 Seismic design

The Site is in a high seismic hazard region. Structures and equipment shall be designed per the Turkish Building Earthquake Code (TBDY 2018) with a site-specific hazard study; design earthquake DD-2 (475-year) peak ground acceleration 0.45 g; building importance factor 1.5 for power block structures; functional recovery of critical systems after DD-3 events.

Discipline: structural | Verification: analysis | Acceptance: Seismic design report approved; site-specific hazard study delivered

ER-02.07 Geotechnical conditions

Owner-provided baseline investigation indicates alluvial deposits over weathered rock at 18 m to 25 m depth, groundwater at +13.5 m and liquefaction potential in the upper sand layers. The Contractor shall perform confirmatory investigations and select foundation solutions (piling or ground improvement) accordingly.

Discipline: civil | Verification: analysis

ER-02.08 Corrosive coastal atmosphere

Outdoor equipment and structures shall be designed for corrosivity category C5 (ISO 12944-2) with a coating system of high durability (H, more than 15 years to first major maintenance).

Discipline: mechanical | Verification: inspection

ER-02.09 Flood protection

Finished floor levels of all buildings and equipment foundations shall be at least 0.5 m above the 100-year coastal flood level including a sea-level-rise allowance of 0.5 m.

Discipline: civil | Verification: analysis

ER-03 Fuel

ER-03.01 Fuel

The only fuel is pipeline natural gas from the national transmission system. No back-up liquid fuel is required; space shall be reserved for a future distillate system.

Discipline: process | Verification: review

ER-03.02 Fuel gas specification

Reference composition (mol %): CH₄ 93.0, C₂H₆ 3.5, C₃H₈ 0.8, C₄+ 0.3, N₂ 1.6, CO₂ 0.8; LHV 48.4 MJ/kg. The plant shall accept the full network range: LHV 45.0 MJ/kg to 50.0 MJ/kg, Wobbe index 47 MJ/Nm³ to 52 MJ/Nm³, H₂S below 5 mg/Nm³, and hydrogen admixture up to 10 vol %.

Discipline: process | Verification: analysis | Acceptance: OEM confirmation of fuel range and H₂ capability

ER-03.03 Gas supply conditions at terminal point

Pressure at the terminal point 45 barg to 70 barg; temperature 5 degC to 25 degC. The gas receiving station shall provide the GT fuel skid with gas at the OEM-required pressure and superheat under all conditions without gas compressors.

Discipline: process | Verification: analysis

ER-03.04 Gas receiving and fiscal metering station

2x100 % filtration/separation, 2x100 % water-bath or electric heaters, 2x100 % pressure-reduction streams with active/monitor regulators and slam-shut valves; fiscal metering with 2x100 % ultrasonic meters (OIML R137 accuracy class 0.5), on-line gas chromatograph and flow computers complying with the network operator's requirements.

Discipline: mechanical | Verification: inspection

ER-03.05 Performance fuel gas heating

The GT fuel gas shall be preheated with intermediate-pressure feedwater to the OEM-optimised temperature (about 200 degC) to maximise combined cycle efficiency; a start-up heater shall be provided.

Discipline: process | Verification: review

ER-04 Performance and operational requirements**ER-04.01 Net electrical output**

Guaranteed net electrical output at SRC: not less than 600.0 MW at the 380 kV terminal point.

Discipline: process | Verification: test | Acceptance: ≥ 600.0 MW corrected to SRC | Guarantee: PG-01

ER-04.02 Net heat rate

Guaranteed net heat rate at SRC: not more than 5,860 kJ/kWh (LHV), corresponding to a net efficiency of about 61.4 %.

Discipline: process | Verification: test | Acceptance: $\leq 5,860$ kJ/kWh corrected to SRC | Guarantee: PG-02

ER-04.03 Part-load efficiency

Net heat rate at 60 % plant load shall not exceed 6,250 kJ/kWh (LHV) at SRC.

Discipline: process | Verification: test | Acceptance: $\leq 6,250$ kJ/kWh | Guarantee: PG-03

ER-04.04 Start-up times

From GT ignition to base load: hot start (shutdown < 8 h) ≤ 45 min; warm start (8 h to 48 h) ≤ 90 min; cold start (> 48 h) ≤ 180 min, without exceeding the life consumption limits agreed with the OEM.

Discipline: process | Verification: demonstration | Acceptance: Hot start ≤ 45 min demonstrated three times | Guarantee: PG-09

ER-04.05 Load ramp rate

Combined cycle load ramp rate of at least 35 MW/min between minimum environmental load and base load.

Discipline: i_and_c | Verification: demonstration | Acceptance: ≥ 35 MW/min | Guarantee: PG-10

ER-04.06 Minimum environmental load

Stable emissions-compliant operation shall be possible down to 40 % of plant net output or lower.

Discipline: process | Verification: demonstration | Acceptance: ≤ 40 % | Guarantee: PG-11

ER-04.07 Frequency and voltage support

The plant shall provide primary and secondary frequency control, fast frequency response and reactive power support as required by the Grid Code, including operation within 47.5 Hz to 52.5 Hz for the durations specified therein.

Discipline: electrical | Verification: test | Acceptance: Grid Code compliance tests accepted by the TSO

ER-04.08 Reactive capability

Each generator shall be capable of power factor 0.85 lagging to 0.95 leading at rated active power at the generator terminals.

Discipline: electrical | Verification: test

ER-04.09 House load and island operation

The unit shall survive a full load rejection to house load and remain in stable operation for at least 60 minutes.

Discipline: i_and_c | Verification: demonstration

ER-04.10 Auxiliary power

Auxiliary power consumption at base load shall not exceed 2.2 % of gross output; the value is included in the net output guarantee.

Discipline: electrical | Verification: test

ER-04.11 Degradation

The Contractor shall submit recoverable and non-recoverable degradation curves for output and heat rate over 48,000 EOH; they apply for test corrections.

Discipline: process | Verification: review

ER-04.12 Availability target

The design shall support an equivalent availability factor of at least 93 % averaged over the first six years, considering the OEM maintenance programme.

Discipline: general | Verification: analysis

ER-05 Gas turbine

ER-05.01 Gas turbine type

Heavy-duty, single-shaft, 50 Hz, H-class (or equivalent) gas turbine with dry low-NOx combustion capable of the fuel range in ER-03.02.

Discipline: mechanical | Verification: review

ER-05.02 Inlet air system

Three-stage static air filtration (coastal environment, final stage EPA E12 per EN 1822), anti-icing, evaporative cooler provision, silencer.

Discipline: mechanical | Verification: inspection

ER-05.03 Combustion dynamics monitoring

On-line combustion dynamics monitoring with automatic tuning shall be provided.

Discipline: i_and_c | Verification: demonstration

ER-05.04 Enclosure and ventilation

Acoustic enclosure with ventilation, gas detection and fire suppression suitable for the hazardous area classification.

Discipline: mechanical | Verification: inspection

ER-05.05 Compressor washing

On-line and off-line compressor washing systems with detergent skid.

Discipline: mechanical | Verification: inspection

ER-05.06 Condition monitoring and remote diagnostics

Vibration, performance and combustion monitoring with secure remote diagnostic access for the OEM (IEC 62443 compliant).

Discipline: i_and_c | Verification: review

ER-05.07 Maintenance intervals

Hot gas path inspection interval shall be not less than 25,000 EOH and major inspection not less than 50,000 EOH at the operating regime of ER-01.03.

Discipline: mechanical | Verification: review

ER-06 Heat recovery steam generator

ER-06.01 HRSG type

Horizontal-gas-flow, triple-pressure, reheat, natural-circulation HRSG, designed for fast cycling (Benson or equivalent HP section acceptable).

Discipline: mechanical | Verification: review

ER-06.02 Design code

Pressure parts per ASME Section I or EN 12952, PED 2014/68/EU conformity assessment by a notified body.

Discipline: mechanical | Verification: inspection | Acceptance: PED certificates delivered

ER-06.03 SCR and CO catalyst

In-duct SCR with aqueous ammonia (< 25 %) injection and an oxidation catalyst sized for the emission guarantees at end of catalyst life (24,000 h).

Discipline: process | Verification: test | Guarantee: PG-04

ER-06.04 Ammonia system

Aqueous ammonia storage (2x7 days at base load), unloading, vapourisation and dosing with leak detection and bunding.

Discipline: process | Verification: inspection

ER-06.05 Cycling capability

The HRSG shall withstand the start regime of ER-01.03 without drum or header fatigue beyond design life (fatigue analysis per EN 12952-3).

Discipline: mechanical | Verification: analysis

ER-06.06 Stack

Steel stack 60 m high (subject to the EIA dispersion study), with CEMS platform, sampling ports per EN 15259 and aviation lighting.

Discipline: structural | Verification: inspection

ER-06.07 Drains and blowdown

Continuous and intermittent blowdown, HRSG drains recovery to the condenser where water quality permits.

Discipline: process | Verification: review

ER-06.08 Preservation

Dry and wet lay-up systems (nitrogen capping and dehumidified air) for long shutdowns.

Discipline: process | Verification: inspection

ER-07 Steam turbine and condenser

ER-07.01 Steam turbine

Reheat condensing steam turbine with HP, IP and LP sections, axial or side exhaust, 50 Hz, with HP/IP/LP bypass systems of 100 % capacity.

Discipline: mechanical | Verification: review

ER-07.02 Main steam conditions

HP live steam about 170 bar(a)/600 degC and hot reheat about 600 degC at base load, as optimised by the OEM.

Discipline: process | Verification: review

ER-07.03 Condenser

Surface condenser with titanium tubes (or super-duplex), divided water boxes for on-load cleaning, ball-type tube cleaning system.

Discipline: mechanical | Verification: inspection

ER-07.04 Cooling water temperature rise

Temperature rise across the condenser shall not exceed 7 K at base load.

Discipline: process | Verification: test | Acceptance: $\leq 7\text{ K}$ | Guarantee: PG-13

ER-07.05 Vacuum system

2x100 % liquid-ring vacuum pumps (holding) plus hogging capability for fast start-up.

Discipline: mechanical | Verification: inspection

ER-07.06 Lube and control oil

Fire-resistant control fluid, lube oil system with 2x100 % AC pumps, DC emergency pump and oil purification.

Discipline: mechanical | Verification: inspection

ER-08 Cooling water system

ER-08.01 Once-through seawater cooling

Offshore submerged intake head with velocity cap (intake velocity $\leq 0.15\text{ m/s}$), intake pipelines, pump house, and outfall with diffuser.

Discipline: civil | Verification: inspection

ER-08.02 Screening

Bar screens, 2x100 % travelling band screens with fish return system, trash rakes.

Discipline: mechanical | Verification: inspection

ER-08.03 Circulating water pumps

2x50 % vertical CW pumps with a common spare pump assembly stored on Site; materials suitable for seawater (duplex).

Discipline: mechanical | Verification: test

ER-08.04 Biofouling control

Electro-chlorination with intermittent/shock dosing; residual chlorine at outfall ≤ 0.1 mg/l.

Discipline: process | Verification: test | Acceptance: ≤ 0.1 mg/l residual chlorine

ER-08.05 Thermal plume

The outfall shall limit the temperature excess to ≤ 3 K at the edge of the 100 m mixing zone.

Discipline: process | Verification: test | Acceptance: ≤ 3 K | Guarantee: PG-14

ER-08.06 Closed cooling water

Closed cooling water circuit for auxiliaries with 2x100 % plate heat exchangers (titanium plates) and 2x100 % pumps.

Discipline: mechanical | Verification: inspection

ER-09 Water treatment and effluents

ER-09.01 Raw water

Seawater reverse osmosis (2x100 %) producing service and demineralisation feed water; no groundwater abstraction.

Discipline: process | Verification: test

ER-09.02 Demineralised water

Two-pass RO plus electrodeionisation, 2x100 %, conductivity < 0.1 uS/cm, silica < 10 ug/l; storage for 3 days of cycle make-up.

Discipline: process | Verification: test

ER-09.03 Cycle chemistry

Water/steam chemistry per VGB-S-010 (all-volatile treatment, oxygenated where applicable) with on-line analysers and sampling panel.

Discipline: process | Verification: review

ER-09.04 Condensate polishing

Full-flow condensate polishing for start-up and seawater ingress events.

Discipline: process | Verification: inspection

ER-09.05 Waste water

Neutralisation, oil separation and treatment so that effluents meet the Turkish Water Pollution Control Regulation discharge limits before the outfall.

Discipline: hse | Verification: test | Acceptance: Effluent within permit limits

ER-09.06 Stormwater

Separate clean and potentially oily stormwater drainage with oil interceptors.

Discipline: civil | Verification: inspection

ER-10 Balance of plant mechanical systems

ER-10.01 Feed water pumps

2x100 % variable-speed (VFD or hydraulic coupling) HP/IP boiler feed pumps.

Discipline: mechanical | Verification: test

ER-10.02 Condensate pumps

2x100 % condensate extraction pumps.

Discipline: mechanical | Verification: test

ER-10.03 Compressed air

2x100 % oil-free compressors, 2x100 % dryers, instrument air dew point -40 degC, receiver for 15 min of instrument air after loss of compressors.

Discipline: mechanical | Verification: test

ER-10.04 Auxiliary boiler

Gas-fired auxiliary boiler for start-up and sealing steam, sized for a cold start without external steam.

Discipline: mechanical | Verification: test

ER-10.05 Fire protection

Fire detection and protection per NFPA 850 and Turkish Fire Regulation: ring main, 2x100 % electric plus diesel fire pumps, water mist/CO2 for GT enclosure, deluge for transformers, gas detection, fire alarm system.

Discipline: hse | Verification: test | Acceptance: Insurer and fire brigade acceptance

ER-10.06 HVAC

HVAC for all buildings; electrical and control rooms with N+1 cooling and positive pressure with filtered (coastal) air.

Discipline: hvac | Verification: test

ER-10.07 Cranes

Turbine hall EOT crane sized for the heaviest maintenance lift (excluding generator stator), and maintenance hoists for pumps and valves.

Discipline: mechanical | Verification: test

ER-10.08 Nitrogen and hydrogen

Nitrogen supply for purging and preservation; hydrogen supply for generator cooling if hydrogen-cooled generators are offered.

Discipline: mechanical | Verification: inspection

ER-11 Electrical systems

ER-11.01 Grid connection

Connection to the 380 kV transmission system via a GIS switchyard with double busbar and bus coupler, two line bays and two generator transformer bays, designed for 50 kA short-circuit current.

Discipline: electrical | Verification: test

ER-11.02 Generators

Air- or hydrogen-cooled synchronous generators per IEC 60034, 50 Hz, static excitation with PSS, generator circuit breakers per IEC/IEEE 62271-37-013.

Discipline: electrical | Verification: test

ER-11.03 Generator step-up transformers

One GSU per generator, ONAN/ONAF, rated for the maximum generator output at -8 degC, on-load tap changer, per IEC 60076; fire walls between transformers.

Discipline: electrical | Verification: test

ER-11.04 Auxiliary power supply

Two unit auxiliary transformers, 10.5 kV MV switchboards (2 sections with bus tie), 0.4 kV PMCC with automatic transfer. Start-up power from the grid via back-feed through the GSU.

Discipline: electrical | Verification: test

ER-11.05 Emergency power

Emergency diesel generator for safe shutdown loads (turning gear, lube oil, emergency lighting); the diesel fire pump has its own independent drive; UPS (2x100 %) for DCS and critical loads with 60 min autonomy; 220 V DC and 24 V DC systems.

Discipline: electrical | Verification: test

ER-11.06 Protection

Duplicated main protection (Main 1/Main 2 from different manufacturers where required by the TSO) for generators, transformers, busbars and lines.

Discipline: electrical | Verification: test

ER-11.07 Earthing and lightning

Earthing grid per IEEE 80 with safe touch and step voltages; lightning protection per IEC 62305.

Discipline: electrical | Verification: test

ER-11.08 Cables

Cables per IEC standards, halogen-free flame-retardant inside buildings, fire-rated cables for safety circuits.

Discipline: electrical | Verification: inspection

ER-11.09 Metering and TSO interface

Revenue metering per TSO/market operator rules (main and check meters, class 0.2S), RTU and telecommunication to the national dispatch centre.

Discipline: electrical | Verification: test

ER-12 Instrumentation, control and cybersecurity

ER-12.01 Distributed control system

Single integrated plant DCS for GT, ST, HRSG and BOP with redundant controllers, networks and power supplies; operation of the whole plant from the central control room by two operators.

Discipline: i_and_c | Verification: test

ER-12.02 Automation level

Automatic start-up and shutdown sequences (one-button start to base load) and automatic load control from the dispatch set-point.

Discipline: i_and_c | Verification: demonstration

ER-12.03 Safety instrumented systems

SIS per IEC 61511 with SIL verification, including burner management and HRSG protection.

Discipline: i_and_c | Verification: analysis | Acceptance: SIL verification reports approved

ER-12.04 CEMS

Certified continuous emission monitoring (NO_x, CO, O₂, NH₃, flow, moisture) per EN 14181 with data transmission to the environmental authority.

Discipline: i_and_c | Verification: test

ER-12.05 Historian and performance monitoring

Plant historian (5 years on-line storage) and on-line performance calculation per ASME PTC 46 principles.

Discipline: i_and_c | Verification: test

ER-12.06 Operator training simulator

High-fidelity operator training simulator of the whole plant, delivered 6 months before first fire.

Discipline: i_and_c | Verification: demonstration

ER-12.07 Cybersecurity

OT cybersecurity per IEC 62443 (target security level SL2), zones and conduits, DMZ, patch and backup management, security acceptance test.

Discipline: i_and_c | Verification: test | Acceptance: Security acceptance test passed

ER-12.08 Time synchronisation

GPS time synchronisation of DCS, protection relays and recorders (1 ms sequence of events).

Discipline: i_and_c | Verification: test

ER-13 Civil, structural and architectural works

ER-13.01 Foundations

Turbine and generator foundations designed for dynamic loads with separated foundations from buildings; settlement limits per OEM.

Discipline: civil | Verification: analysis

ER-13.02 Buildings

Turbine hall, control and electrical building, administration building, workshop/warehouse, water treatment building, gatehouse, fire station.

Discipline: civil | Verification: inspection

ER-13.03 Structural steel

Structural steel per EN 1993 with seismic detailing, hot-dip galvanised outdoors (EN ISO 1461).

Discipline: structural | Verification: inspection

ER-13.04 Concrete

Concrete per EN 1992 / TS 500 with exposure class XS for marine-exposed elements; minimum 50-year durability.

Discipline: civil | Verification: inspection

ER-13.05 Roads and drainage

Internal asphalt roads for 40 t axle loads to the heaviest delivery route, surface water drainage for ER-02.05 rainfall.

Discipline: civil | Verification: inspection

ER-13.06 Marine works

Intake and outfall pipelines (HDPE or GRP) with ballast and scour protection, designed for 50-year wave conditions.

Discipline: civil | Verification: analysis

ER-14 Environment, health and safety

ER-14.01 EIA compliance

The Works shall comply with the conditions of the positive EIA decision and the Environmental Law (No. 2872) and its regulations.

Discipline: hse | Verification: review | Acceptance: No outstanding EIA non-conformities at Taking-Over

ER-14.02 Air emissions

NO_x ≤ 30 mg/Nm³ and CO ≤ 30 mg/Nm³ (dry, 15 % O₂) between 50 % and 100 % GT load; NH₃ slip ≤ 5 mg/Nm³.

Discipline: hse | Verification: test | Acceptance: Within guaranteed limits | Guarantee: PG-04

ER-14.03 Near-field noise

Maximum 85 dB(A) at 1 m from any equipment in normal operation.

Discipline: hse | Verification: test | Acceptance: ≤ 85 dB(A) | Guarantee: PG-07

ER-14.04 Environmental noise

Maximum 45 dB(A) LAeq night at the nearest sensitive receptor.

Discipline: hse | Verification: test | Acceptance: <= 45 dB(A) | Guarantee: PG-08

ER-14.05 Occupational health and safety

Construction and design shall comply with the Occupational Health and Safety Law (No. 6331); the Contractor shall implement an ISO 45001-certified HSE management system and a lost-time-injury target of zero.

Discipline: hse | Verification: review

ER-14.06 Hazardous areas

Hazardous area classification per IEC 60079-10-1; equipment certified accordingly.

Discipline: hse | Verification: review

ER-14.07 Process safety studies

HAZID, HAZOP, SIL determination, fire and explosion risk assessment and a Major Accident Prevention assessment where applicable.

Discipline: hse | Verification: review | Acceptance: All actions closed before first fire

ER-14.08 Carbon capture readiness

Space reservation and tie-in points for a future post-combustion carbon capture plant at the HRSG outlet.

Discipline: process | Verification: review

ER-15 Quality, inspection and testing during manufacture

ER-15.01 Quality management

ISO 9001 quality management for the Contractor and all major subcontractors; project quality plan within 30 days of NTP.

Discipline: general | Verification: review

ER-15.02 Inspection and test plans

ITPs for all major equipment with hold and witness points for the Engineer, 15 days' notice for witnessed tests.

Discipline: general | Verification: review

ER-15.03 Factory acceptance tests

FATs at least for: GT and ST control systems, generators, GSU transformers, GIS, MV switchgear, DCS, CEMS.

Discipline: general | Verification: test

ER-15.04 Material traceability

Material certificates per EN 10204 3.1 (3.2 for pressure parts above 50 bar) and positive material identification of alloy piping.

Discipline: mechanical | Verification: inspection

ER-15.05 Welding

Welding per ASME IX / EN ISO 15614, 100 % NDE of HP/HRH/main steam butt welds.

Discipline: piping | Verification: inspection

ER-16 Commissioning and Tests on Completion

ER-16.01 Commissioning plan

System-based commissioning plan with pre-commissioning, cold and hot commissioning, submitted 9 months before first fire.

Discipline: commissioning | Verification: review

ER-16.02 Steam blowing and chemical cleaning

Chemical cleaning of the HRSG and steam blowing of steam lines to OEM cleanliness criteria.

Discipline: commissioning | Verification: test

ER-16.03 Grid Code compliance tests

All tests required by the TSO for connection and commercial operation, including reactive capability and frequency response.

Discipline: electrical | Verification: test | Acceptance: TSO acceptance

ER-16.04 Performance test

Performance test per ASME PTC 46 within 30 days after initial operation, test procedure submitted 90 days before, with instrument calibration traceable to national standards.

Discipline: commissioning | Verification: test | Acceptance: PG-01/PG-02/PG-03 demonstrated | Guarantee: PG-01

ER-16.05 Emission and noise tests

Emission tests by an accredited laboratory and noise surveys per PG-04 to PG-08.

Discipline: hse | Verification: test | Guarantee: PG-04

ER-16.06 Operational demonstration

Demonstration of start times, ramp rates, minimum load, load rejection and trip tests.

Discipline: commissioning | Verification: demonstration | Guarantee: PG-09

ER-16.07 Reliability run

30-day continuous reliability run with availability $\geq 98\%$; restarted after rectification if failed.

Discipline: commissioning | Verification: test | Acceptance: $\geq 98\%$ | Guarantee: PG-12

ER-17 Documentation, training, O&M and spares

ER-17.01 Document management

Documents submitted electronically with the Owner's numbering and a master document register; review period 14 days (21 days for the Basic Design Package).

Discipline: project_control | Verification: review

ER-17.02 Schedule and reporting

Level 3 CPM schedule (MS Project XML or Primavera P6) with monthly updates, S-curves and progress reports.

Discipline: project_control | Verification: review

ER-17.03 As-built documentation

As-built drawings, 3D model and engineering database within 3 months of Taking-Over.

Discipline: general | Verification: review

ER-17.04 O&M manuals

Operation and maintenance manuals (English, operating procedures also in Turkish) 6 months before first fire, final at Taking-Over.

Discipline: general | Verification: review

ER-17.05 Training

Classroom, simulator and on-the-job training for 60 Owner staff (operators, maintenance, I&C, chemistry), minimum 12 weeks per key trade.

Discipline: general | Verification: demonstration

ER-17.06 Spare parts

Commissioning spares, special tools, and a recommended two-year operating spares list with prices valid for 18 months.

Discipline: general | Verification: review

ER-17.07 Long-term service

The power island supplier shall offer a long-term service agreement for GT/ST/generators (separate contract, not part of the Contract Price).

Discipline: general | Verification: review

ER-18 Interfaces and terminal points

ER-18.01 Terminal points

Terminal points and scope limits are defined in Appendix D; the Contractor shall provide all works up to and including the terminal points.

Discipline: general | Verification: review

ER-18.02 Grid connection interface

The Contractor shall coordinate with the TSO and the Owner's transmission line contractor for protection settings, SCADA and energisation.

Discipline: electrical | Verification: review

ER-18.03 Gas network interface

The Contractor shall coordinate the gas receiving station tie-in, pressure protection and metering approval with the gas network operator.

Discipline: process | Verification: review

ER-18.04 Owner's engineering database

All interface data shall be exchanged through the Owner's engineering database (single source of truth).

Discipline: general | Verification: review

Appendix B - Performance Guarantees and Liquidated Damages

ID	Guarantee	Value	Conditions	Test	Liquidated damages / remedy	Minimum acceptance
PG-01	Net electrical output at Site Reference Conditions	>= 600 MW	Base load, new and clean (corrected for recoverable degradation per agreed curves), Site Reference Conditions (ER-02.03), power factor 0.85 lagging at the 380 kV terminal point, all auxiliaries in normal operation.	ASME PTC 46 overall plant performance test, corrections to Site Reference Conditions per agreed correction curves; measured at the 380 kV revenue meters.	2,200 EUR per kW shortfall. Below 97 % of the guaranteed value the Owner may reject the Works after the rectification period (Sub-Clause 12.5).	582 MW
PG-02	Net heat rate (LHV) at Site Reference Conditions	<= 5860 kJ/kWh	As PG-01, fuel within the Fuel Specification (ER-03), heat input on LHV basis.	ASME PTC 46, fuel flow by fiscal ultrasonic meters, LHV by on-line gas chromatograph (ISO 6976).	400,000 EUR per kJ/kWh excess. Above 103 % of the guaranteed value the Owner may reject the Works after the rectification period.	6035.8 kJ/kWh
PG-03	Net heat rate (LHV) at 60 % plant load	<= 6250 kJ/kWh	Plant at 60 % of net output, Site Reference Conditions, sliding pressure.	ASME PTC 46, part-load test point.	150,000 EUR per kJ/kWh excess	-
PG-04	NOx emission (as NO2)	<= 30 mg/Nm3	Dry flue gas, 15 % O2, 50 % to 100 % GT load, all ambient conditions within the design range.	EN 14792 via certified CEMS (EN 14181 QAL2) plus independent stack test.	No LD: the Contractor shall rectify at its cost until compliant; Taking-Over is not issued before compliance.	-
PG-05	CO emission	<= 30 mg/Nm3	Dry flue gas, 15 % O2, 50 % to 100 % GT load.	EN 15058 via CEMS plus stack test.	Must rectify (as PG-04).	-
PG-06	Ammonia slip from the SCR	<= 5 mg/Nm3	Dry flue gas, 15 % O2, 50 % to 100 % GT load, catalyst new.	Stack test per recognised method (e.g. EN ISO 21877).	Must rectify.	-
PG-07	Near-field noise	<= 85 dB(A)	1 m from equipment surfaces / enclosures, 1.5 m height, all operating modes except emergency.	ISO 3744 / ISO 3746 survey.	Must rectify.	-
PG-08	Noise at nearest sensitive receptor (night)	<= 45 dB(A)	L _{Aeq} at the nearest residential receptor defined in the EIA, plant at base load, night period.	Environmental noise measurement per ISO 1996-2 with background correction.	250,000 EUR per dB(A) exceedance (after rectification attempts). Contractor to rectify first; LD only if rectification fails.	-
PG-09	Hot start time to base load	<= 45 min	Shutdown shorter than 8 h, GT ignition to plant base load, emissions compliant from minimum environmental load.	Demonstration during Tests on Completion, three successful starts.	20,000 EUR per minute exceedance	-
PG-10	Load ramp rate in combined cycle	>= 35 MW/min	Between minimum environmental load and base load, emissions compliant.	Demonstration, recorded by DCS historian.	100,000 EUR per MW/min shortfall	-
PG-11	Minimum environmental load	<= 40 % of plant net output	Lowest load with PG-04/PG-05 compliance at Site Reference Conditions.	Demonstration with CEMS records.	300,000 EUR per percentage point exceedance	-
PG-12	Reliability run availability	>= 98 %	30 consecutive days of operation following Owner dispatch, maximum two forced outages, none longer than 48 h.	Reliability Run per ER-16.07.	Failed run to be repeated from the start after rectification.	-
PG-13	Condenser cooling water temperature rise	<= 7 K	Base load, all seawater temperatures within the design range.	Temperature measurement at condenser inlet/outlet.	Must rectify (EIA permit condition).	-
PG-14	Thermal plume at edge of mixing zone	<= 3 K	Temperature excess over ambient seawater at 100 m from the outfall diffuser, base load, summer conditions.	Thermal plume survey per EIA monitoring plan.	Must rectify (EIA permit condition).	-

Caps: performance liquidated damages 15 % and aggregate liquidated damages 20 % of the Contract Price (Sub-Clause 12.6). Delay damages per Sub-Clause 9.5.

Appendix C - Schedule of Payments

ID	Milestone	Payment trigger	Planned month after NTP	Share	Amount	Security
MS-01	Advance payment	Contract effective and receipt of the Advance Payment Guarantee	0	10 %	EUR 51,500,000	Advance Payment Guarantee 10 %
MS-02	Approval of basic design and power island order	Owner approval of the Basic Design Package and evidence of firm power island sub-orders	3	8 %	EUR 41,200,000	
MS-03	Gas turbine and generator manufacturing progress	Evidence of GT rotor forging and generator stator core completion (inspection report)	9	12 %	EUR 61,800,000	
MS-04	Main civil works started	Completion of GT/ST foundation piling and first mat pour	8	5 %	EUR 25,750,000	
MS-05	Delivery of gas turbine to Site	GT package delivered to Site, receiving inspection passed	16	12 %	EUR 61,800,000	
MS-06	Delivery of steam turbine, generators and GSU transformers	Main components delivered to Site, receiving inspection passed	18	10 %	EUR 51,500,000	
MS-07	HRSB pressure parts erected and hydrotested	Successful HRSB hydrostatic test witnessed by the Engineer	20	8 %	EUR 41,200,000	
MS-08	Back-feed energisation	380 kV switchyard and GSU energised from the grid, auxiliary power available	21	5 %	EUR 25,750,000	
MS-09	First fire of the gas turbine	GT first fire on natural gas	23	7 %	EUR 36,050,000	
MS-10	First synchronisation in combined cycle	Steam turbine synchronised to the grid	26	8 %	EUR 41,200,000	
MS-11	Taking-Over of the Works	Issue of the Taking-Over Certificate (after Tests on Completion and Reliability Run)	32	10 %	EUR 51,500,000	Retention release against Retention Guarantee
MS-12	Final completion	Issue of the Performance Certificate at the end of the Defects Notification Period	56	5 %	EUR 25,750,000	
	Total			100 %	EUR 515,000,000	

Amounts are computed from the Contract Price; the advance is recovered pro rata from MS-02 to MS-10 and 5 % retention applies to each milestone (Clause 15).

Appendix D - Scope of Works Split and Terminal Points

Party codes: IEC = Imaginary Electric Company; IEPC = Istanbul EPC Mühendislik ve Taahhüt A.Ş.; OWNER = Alpilla Enerji Üretim A.Ş.

ID	Area	Item	Design	Supply	Install	Commission	Terminal point
SC-001	Site	Land, access rights and site handover (levelled to +15.0 m)	OWNER	OWNER	OWNER	-	Site boundary per land-use plan
SC-002	Site	Environmental Impact Assessment (EIA) positive decision	OWNER	OWNER	-	-	
SC-003	Site	Construction, operating and building permits (Contractor supplies all design inputs)	IEPC	OWNER	-	-	

ID	Area	Item	Design	Supply	Install	Commission	Terminal point
SC-004	Site	Geotechnical and topographic surveys (baseline supplied by Owner, confirmatory by Contractor)	IEPC	IEPC	IEPC	-	
SC-005	Site	Site preparation, fencing, roads, drainage, temporary facilities	IEPC	IEPC	IEPC	-	
SC-006	Power island	Gas turbine package incl. auxiliaries, enclosure, inlet air filtration, exhaust diffuser	IEC	IEC	IEPC	IEC	GT exhaust flange to HRSG inlet duct
SC-007	Power island	GT generator incl. excitation, protection, cooling	IEC	IEC	IEPC	IEC	Generator terminals (IPB connection)
SC-008	Power island	Steam turbine (HP/IP/LP, reheat) incl. lube/control oil, gland steam, turning gear	IEC	IEC	IEPC	IEC	
SC-009	Power island	ST generator incl. excitation, protection	IEC	IEC	IEPC	IEC	Generator terminals
SC-010	Power island	HRSG triple-pressure reheat, horizontal, natural circulation, with SCR and CO catalyst	IEC	IEC	IEPC	IEC	HRSG outlet to stack
SC-011	Power island	Stack 60 m with CEMS platform and silencer	IEPC	IEPC	IEPC	IEPC	
SC-012	Power island	Water-cooled condenser (seawater), vacuum system	IEC	IEC	IEPC	IEC	Cooling water flanges at condenser water boxes
SC-013	Power island	Power island control system (TCS/ST controls) integrated in plant DCS	IEC	IEC	IEPC	IEC	DCS interface (redundant fibre)
SC-014	Power island	GSU transformers (GT and ST), isolated phase busducts, generator circuit breakers	IEC	IEC	IEPC	IEC	GSU HV bushings
SC-015	Balance of plant	Condensate and boiler feed water systems (pumps 2x100 %)	IEPC	IEPC	IEPC	IEPC	
SC-016	Balance of plant	Seawater intake, screens, CW pumps (2x50 %), outfall and diffuser	IEPC	IEPC	IEPC	IEPC	Offshore intake head / outfall diffuser
SC-017	Balance of plant	Electro-chlorination for biofouling control	IEPC	IEPC	IEPC	IEPC	
SC-018	Balance of plant	Seawater reverse osmosis and demineralisation plant	IEPC	IEPC	IEPC	IEPC	
SC-019	Balance of plant	Condensate polishing plant	IEPC	IEPC	IEPC	IEPC	
SC-020	Balance of plant	Waste water treatment and effluent monitoring	IEPC	IEPC	IEPC	IEPC	Discharge point to outfall
SC-021	Balance of plant	Closed cooling water system with plate heat exchangers	IEPC	IEPC	IEPC	IEPC	
SC-022	Balance of plant	Compressed air (service and instrument) 2x100 %	IEPC	IEPC	IEPC	IEPC	
SC-023	Balance of plant	Auxiliary boiler (gas-fired) for start-up steam	IEPC	IEPC	IEPC	IEPC	
SC-024	Balance of plant	Fire detection and protection per NFPA 850	IEPC	IEPC	IEPC	IEPC	
SC-025	Balance of plant	HVAC for all buildings and electrical rooms	IEPC	IEPC	IEPC	IEPC	
SC-026	Balance of plant	Cranes and hoists (turbine hall EOT crane)	IEPC	IEPC	IEPC	IEPC	
SC-027	Fuel gas	Gas pipeline from national transmission system to plant boundary	OWNER	OWNER	OWNER	OWNER	Owner valve at plant boundary (inlet flange of gas receiving station)
SC-028	Fuel gas	Gas receiving and metering station (fiscal metering, filtration, heating, pressure control)	IEPC	IEPC	IEPC	IEPC	Inlet flange at plant boundary
SC-029	Fuel gas	Performance gas heater and GT fuel skid	IEC	IEC	IEPC	IEC	
SC-030	Electrical	380 kV GIS switchyard (double busbar) within plant boundary	IEPC	IEPC	IEPC	IEPC	380 kV line gantry / cable sealing ends
SC-031	Electrical	380 kV transmission connection line and remote substation works	OWNER	OWNER	OWNER	OWNER	Line gantry at switchyard
SC-032	Electrical	Unit auxiliary transformers, MV switchgear 10.5 kV, LV 0.4 kV distribution	IEPC	IEPC	IEPC	IEPC	
SC-033	Electrical	Emergency diesel generator and UPS / DC systems	IEPC	IEPC	IEPC	IEPC	
SC-034	Electrical	Earthing, lightning protection, cabling, cable trays	IEPC	IEPC	IEPC	IEPC	
SC-035	Electrical	Revenue metering and TSO SCADA/telecom interface	IEPC	IEPC	IEPC	IEPC	TSO RTU interface in control building
SC-036	I&C	Plant DCS, historian, operator training simulator	IEPC	IEPC	IEPC	IEPC	
SC-037	I&C	Safety instrumented systems (burner management, HRSG	IEC	IEC	IEPC	IEC	

ID	Area	Item	Design	Supply	Install	Commission	Terminal point
		protection) per IEC 61511					
SC-038	I&C	Continuous emission monitoring system (CEMS)	IEPC	IEPC	IEPC	IEPC	Data link to environmental authority
SC-039	I&C	Plant security, CCTV, access control, cybersecurity (IEC 62443)	IEPC	IEPC	IEPC	IEPC	
SC-040	Civil	Piling, foundations, structures, turbine hall, HRSG structure	IEPC	IEPC	IEPC	-	
SC-041	Civil	Buildings: control, administration, workshop/warehouse, water treatment, gatehouse	IEPC	IEPC	IEPC	-	
SC-042	Civil	Marine works for intake and outfall (dredging, pipes, heads)	IEPC	IEPC	IEPC	-	
SC-043	Utilities	Potable water and sewage connections at site boundary	OWNER	OWNER	OWNER	-	Site boundary
SC-044	Utilities	Telecommunication connection (fibre) to site boundary	OWNER	OWNER	OWNER	-	Site boundary
SC-045	Utilities	Construction power and water	IEPC	IEPC	IEPC	-	
SC-046	Commissioning	Consumables until Taking-Over (except natural gas)	IEPC	IEPC	-	-	
SC-047	Commissioning	Natural gas for commissioning and tests	OWNER	OWNER	-	-	Gas receiving station inlet
SC-048	Commissioning	Grid access for back-feed and test energy; purchase of test energy	OWNER	OWNER	-	-	
SC-049	Commissioning	Operation and maintenance staff for training and commissioning participation	OWNER	OWNER	-	-	
SC-050	Spares	Commissioning spares and special tools	IEPC	IEPC	-	-	
SC-051	Spares	Two-year operating spares (priced list, Owner option)	IEPC	IEPC	-	-	Owner's warehouse

Appendix E - Site and Design Data

Category	Parameter	Value	Unit	Condition	Basis
ambient	Ambient temperature - minimum for full capability	-8	degC		ER-02.02
ambient	Ambient temperature - maximum for full capability	40	degC		ER-02.02
ambient	Ambient temperature - survival range	-15 to +45	degC		ER-02.02
ambient	SRC ambient dry-bulb temperature	15	degC	Site Reference Conditions	ER-02.03
ambient	SRC relative humidity	70	%	Site Reference Conditions	ER-02.03
ambient	SRC barometric pressure	1011.5	mbar	Site Reference Conditions	ER-02.03
emissions	NOx (as NO2)	30	mg/Nm3	dry, 15 % O2, 50-100 % GT load	ER-14.02
emissions	CO	30	mg/Nm3	dry, 15 % O2, 50-100 % GT load	ER-14.02
emissions	NH3 slip	5	mg/Nm3	dry, 15 % O2	ER-14.02
emissions	Stack height	60	m	subject to EIA dispersion study	ER-06.06
fuel gas	Reference fuel LHV	48.4	MJ/kg	reference composition	ER-03.02
fuel gas	Fuel LHV range	45.0 to 50.0	MJ/kg		ER-03.02
fuel gas	Wobbe index range	47 to 52	MJ/Nm3		ER-03.02
fuel gas	Hydrogen admixture capability	10	vol %	maximum	ER-03.02
fuel gas	Gas supply pressure at terminal point	45 to 70	barg		ER-03.03
fuel gas	Gas supply temperature at terminal point	5 to 25	degC		ER-03.03
grid	Grid connection voltage	380	kV	nominal	ER-11.01
grid	Switchyard short-circuit rating	50	kA		ER-11.01
grid	Grid frequency	50	Hz	operating range 47.5 to 52.5 Hz per Grid Code	ER-04.07
grid	Generator power factor range	0.85 lagging to 0.95	-	at rated active power	ER-04.08

Category	Parameter	Value	Unit	Condition	Basis
		leading			
noise	Near-field noise	85	dB(A)	1 m from equipment	ER-14.03
noise	Noise at nearest receptor (night)	45	dB(A)	LAeq night	ER-14.04
operation	Design life	30	years		ER-01.03
operation	Operating hours per year	8000	EOH/year		ER-01.03
operation	Starts per year	250	starts/year	50 cold, 100 warm, 100 hot	ER-01.03
operation	Hot start time to base load	45	min	maximum	ER-04.04
operation	Warm start time to base load	90	min	maximum	ER-04.04
operation	Cold start time to base load	180	min	maximum	ER-04.04
performance	Net electrical output (guaranteed)	600	MW	SRC, 380 kV terminal point	ER-04.01
performance	Net heat rate LHV (guaranteed)	5860	kJ/kWh	SRC	ER-04.02
performance	Net heat rate LHV at 60 % load (guaranteed)	6250	kJ/kWh	SRC	ER-04.03
performance	Auxiliary power consumption	2.2	% of gross	maximum, base load	ER-04.10
performance	Load ramp rate	35	MW/min	minimum	ER-04.05
performance	Minimum environmental load	40	% net output	maximum	ER-04.06
site	Site grade level	15	m a.s.l.		ER-02.01
site	Seismic design PGA (DD-2, 475-year)	0.45	g	TBDY 2018, site-specific hazard study	ER-02.06
site	Atmospheric corrosivity category	C5 (ISO 12944-2), durability H	-		ER-02.08
site	Basic wind speed	30	m/s	10 min mean, 10 m, 50-year return	ER-02.05
site	Ground snow load	0.75	kN/m2		ER-02.05
site	Design rainfall intensity	110	mm/h	10 min, 25-year return	ER-02.05
water	Seawater temperature range at intake	6 to 27	degC		ER-02.04
water	SRC seawater temperature	16	degC	Site Reference Conditions	ER-02.03
water	Condenser cooling water temperature rise	7	K	maximum, base load	ER-07.04
water	Thermal plume excess at 100 m mixing zone	3	K	maximum	ER-08.05
water	Residual chlorine at outfall	0.1	mg/l	maximum	ER-08.04

Appendix F - Parties and Addresses for Notices

Code	Name	Role	Country	Representative	Address	Fictional
IEC	Imaginary Electric Company	consortium member	Germany	Power Island Package Manager	Fictional power-equipment manufacturer	yes
IEPC	Istanbul EPC Mühendislik ve Taahhüt A.Ş.	consortium leader	Türkiye	Consortium Project Director	Istanbul, Türkiye	yes
OE	Owner's Engineer (to be appointed)	owners engineer	Türkiye			yes
OWNER	Alpilla Enerji Üretim A.Ş.	owner	Türkiye	Project Director (Owner)	Alpilla Power Plant Site Office, Marmara coastal industrial zone, Türkiye	yes
TSO	National Transmission System Operator	tso	Türkiye			no

Appendix G - Key Dates

ID	Key date	Date	Obligation
KD-000	Notice to Proceed	2026-11-02	start no earlier than
KD-010	Back-feed energisation (key date)	2028-08-02	complete no later than
KD-020	GT first fire (key date)	2028-10-02	complete no later than
KD-030	First synchronisation in combined cycle (key date)	2029-01-02	complete no later than
KD-040	Taking-Over of the Works (Time for Completion, 975 days)	2029-07-04	complete no later than

Key dates are contractual; the Contractor's Level 3 programme (Sub-Clause 9.3) shall be linked to them.